Appendix: Data Sources and Robustness Checks

Does AI Compress Wages? — Sheng, Zhang & Kandemir (2026)

This appendix reports 13 robustness checks across four categories (Table A0). Category B (Identification) tests functional form of the pre-existing trend, the smoking-gun placebo, and multi-index horse races. Category C (Sample) drops COVID years, excludes the tech sector, and restricts to a balanced panel. Category D (Estimation) applies two-way clustering. Category E (External Validation) provides direct inequality tests, a 72-month employment event study, and individual-level ACS validation.

Table A0: Summary of All Robustness Checks

Cat.	Test	Result	Verdict	Table
B	Linear trend-break (main)	$\delta_2 = +0.0031^{***}$	✓ Holds	B1
B	Quadratic trend-break	$+0.0026^{**}$	✓ Holds	B1
B	Cubic trend-break	$+0.0048^{***}$	✓ Holds	B1
B	Routine intensity smoking gun	DID $+0.83\%^{***} \rightarrow$ TB -0.07% (n.s.)	⇒ Motivates TB	B2
B	Horse race: AIOE vs GPT	Both sig individually ($r = 0.90$)	✓ Same channel	B3
B	Horse race: all 4 indices	n.s. jointly (multicollinearity)	⇒ Collinearity	B3
C	Drop COVID years (2020–21)	$+0.0040^{***}$ (stronger)	✓ Holds	C1
C	Exclude tech sector (soc2≠15)	$+0.0031^{***}$ (unchanged)	✓ Holds	C2
C	Balanced panel	$+0.0026^{**}$ (wage ✓; emp note)	✓ Holds	C3
D	Two-way clustering (occ + state)	$p = 0.014$ (SE: $0.0008 \rightarrow 0.0012$)	✓ Holds*	D1
E	Direct inequality: log(p90/p10)	$\delta_2 = -0.0054^{***}$	✓ Holds	E1
E	CPS 72-month employment event study	No structural break post-ChatGPT	✓ Holds	E2
E	ACS (200K raw → 164,056 matched)	δ_2(p10) is 5.4× δ_2(p90)	✓ Directional	E3

Green = clearly holds; yellow = holds with caveats; ⇒ = motivates correction or note. Outcome: log median wage unless noted.

A. Data Sources

Table A1 lists all datasets used across the occupation-level, firm-level, and validation analyses. All regression tables report the actual N obs used after crosswalk matching.

Table A1: Complete Data Sources

Dataset	Role	N obs	Years	Key variables / Source file
Treatment Variables				
AIOE (Felten et al. 2021)	Main treatment — all occ-level regressions	774 occs	Static	`aioe` (z-scored) · `AIOE_DataAppendix.xlsx`
GPT Exposure (Eloundou et al. 2024)	Placebo tests + horse race	798 occs (663 merged)	Static	$\sum(\gamma \times \text{cw}) / \sum \text{cw}$ · `full_labelset.tsv` + `full_onet_data.tsv`
Frey & Osborne (2013)	Placebo tests + horse race	702 occs	Static	`prob` (automation prob., avg 0.54) · `jobautomation-probability.csv`
Webb AI Index (2020)	Placebo tests + horse race	341 occ codes	Static	`pct_ai` · `exposure_by_occ1990dd.txt` + `onet_to_occ1990dd.txt`
AEI — Anthropic Economic Index	Firm-level Bartik treatment (Sec. 6)	756 occs	Nov 2025	`observed_exposure` · `job_exposure.csv`
AEI Raw Task Data	Firm-level mechanism variables	458,778 task rows	Nov 2025	`ai_autonomy_mean`, `human_only_pct` · `aei_raw_claude_ai_...csv`
Outcome Variables				
BLS OEWS State Panel	Main regression outcomes	157,677 obs	2019–2024	`log_wage`, `log_emp`, `aioe`, `t`, `post`, O*NET scores · `state_panel_final.csv`
BLS OEWS State Percentile Files	Percentile regression + p90/p10 analysis	6 annual files	2019–2024	`a_pct10/25/50/75/90` by state×occ · `state_M20XX_dl.xlsx`

(Table A1 continued)

Dataset	Role	N obs	Years	Key variables / Source file
Compustat North America	Firm-level outcomes	24,845 firm×yr (5,417 firms)	2019–2024	`emp`, `xlr`, `revt`, `at`, `naics` · WRDS
BLS CPS Monthly Employment Panel	Employment event study (Table E2)	32,985 → 25,114 after merge	2019–2024	`emp_rate`, `OCC_CODE` · `cps_emp_monthly.csv`
ACS Individual Wages	Directional floor-raiser validation (Table E3)	200,000 → 164,056 after merge	2019, 2021–2024	`LOG_WAGE`, `OCC`, `PERWT` · `acs_indiv_small.csv`
Crosswalks				
O*NET-to-occ1990dd Crosswalk	Webb: O*NET SOC → Census 1990 codes	O*NET occs	Static	`onetsoccode`, `occ1990dd` · `onet_to_occ1990dd.txt`
BLS CPS-to-NEM Crosswalk	CPS/ACS: Census OCC integer codes → SOC	525 mappings	Aug 2025	`cps_code` → `nem_code` (SOC) · `nem-occcode-cps-crosswalk.xlsx`
BLS Industry-Occupation Matrix	AEI → Compustat: Bartik shift-share weights	231 NAICS industries	2019 baseline	Employment shares → `industry_ai_exposure` (Bartik) · `national_M2022_dl.xlsx`

'Static' = single cross-section merged as time-invariant variable to the 2019–2024 panel. After merge: 157,677 obs for main occupation regressions; 25,114 obs for CPS event study (76.1% match); 164,056 obs for ACS validation (82.0% match); 24,845 firm-year obs for Compustat.

A.1 Treatment Variable Construction

AIOE: Z-scored Felten et al. (2021) index pre-merged in `state_panel_final.csv` by SOC code. All interaction terms pre-computed (`aioe_t` = `aioe` × `t`; `aioe_post` = `aioe`

$\times$ `post`).

GPT Exposure: $\text{GPT_exposure}_o = \sum_k(\gamma_k \times \text{coreweight}_k) \,/\, \sum_k \text{coreweight}_k$, where $\gamma = 1$ if GPT-4 with full tools can perform task k. Built from `full_labelset.tsv` + `full_onet_data.tsv`. Yields 798 occupations, 663 merging to BLS panel. Z-scored. $r(\text{AIOE}) = 0.901$.

Frey & Osborne: Automation risk probability from `jobautomationprobability.csv`, merged by SOC code. Z-scored. $r(\text{AIOE}) = -0.424$. Captures industrial-era substitution risk, not LLM exposure.

Webb AI Index: Patent text similarity at Census 1990 occupation level. Two-step: `onet_to_occ1990dd.txt` maps O*NET SOC $\rightarrow$ occ1990dd codes; averaged back to 6-digit SOC. Z-scored. $r(\text{AIOE}) = 0.202$.

AEI Bartik:

$$\text{industry_ai_exposure}_j = \sum_k \left[\frac{\text{emp}(j,k)}{\sum_k \text{emp}(j,k)} \right] \times \text{observed_exposure}_k$$

using `national_M2022_dl.xlsx` employment shares. Highest industries: Business Support Services (0.430), Securities (0.355), Software Publishers (0.318).

A.2 Crosswalk Paths

Path 1 — Occupation-level
(AIOE/GPT/F&O/Webb $\rightarrow$ BLS OEWS): AIOE and O*NET scores pre-merged in `state_panel_final.csv` by SOC. GPT Exposure aggregated to 6-digit SOC. F&O merged by SOC directly. Webb: two-step via `onet_to_occ1990dd.txt`, then average to SOC.

Path 2 — Firm-level (AEI $\rightarrow$ Compustat): Compustat firms identified by 4-digit NAICS. Bartik shift-share using `national_M2022_dl.xlsx` produces 231 industry-level AEI scores, merged to Compustat by NAICS.

Path 3 — CPS/ACS (Census OCC $\rightarrow$ SOC): Both CPS Monthly Panel and ACS use integer Census codes. BLS CPS-to-NEM crosswalk (`nem-occcode-cps-crosswalk.xlsx`, Aug 2025; 525 mappings) converts to SOC. Match rates: 76.1% for CPS monthly (351/474 occs), 82.0% for ACS (443/529 occs).

B. Identification Robustness

Category B tests whether the main result is sensitive to the assumption that the pre-existing differential trend is linear, and whether the treatment variable captures a genuine AI channel. The causal estimate δ_2 remains positive and significant across linear, quadratic, and cubic trend specifications.

Table B1: Quadratic and Cubic Trend-Break DID — Log Median Wage

	Linear (main)	**Quadratic**	**Cubic**
δ_1 (AIOE$\times t$)	-0.0078^{***}	-0.0080^{***}	-0.0021^{*}
SE	(0.0003)	(0.0005)	(0.0009)
δ_2 (AIOE$\times t^2$)	—	$+0.0001$ n.s.	-0.0032^{***}
SE	—	(0.0001)	(0.0005)
δ_3 (AIOE$\times t^3$)	—	—	$+0.0004^{***}$
SE	—	—	(0.0001)
δ (AIOE$\times$Post) ← causal	$+0.0031^{***}$	$+0.0026^{**}$	$+0.0048^{***}$
SE	(0.0008)	(0.0008)	(0.0009)
N obs	**157,677**	**157,677**	**157,677**
FE: Occ×State + Year + State×Year | SE clustered by occ×state			

*** $p < 0.001$ ** $p < 0.01$ * $p < 0.05$ n.s. = not significant. Causal estimate (green row) remains positive and significant across all functional forms: $+0.0031^{***}$ (linear), $+0.0026^{**}$ (quadratic), $+0.0048^{***}$ (cubic).

B2. Routine Intensity Smoking Gun

Routine task intensity is a pre-AI occupational characteristic with no direct ChatGPT mechanism. It exhibits a pre-existing wage trend due to job polarization dynamics. Standard DID produces a spuriously significant β; Trend-Break DID reduces it to zero — mechanically demonstrating the bias.

Table B2: Routine Intensity Smoking Gun — Log Median Wage

*** $p < 0.001$ n.s. = not significant. Standard DID produces $\beta = +0.83\%^{***}$ for a variable with no ChatGPT mechanism. Trend-Break DID reduces this to $\delta_2 = -0.07\%$ ($p = 0.427$). Contrast with AIOE ($\delta_2 = +0.31\%^{***}$) demonstrates the bias mechanically. Red panel = biased; green panel = corrected.

	Routine Intensity	AIOE (reference)
Panel A: Standard DID — BIASED		
β (Index×Post)	$+0.0083^{***}$	-0.0201^{***}
SE	(0.0007)	(0.0006)
Panel B: Trend-Break DID — CORRECTED		
δ_1 (Index×t)	$+0.0030^{***}$	-0.0078^{***}
SE	(0.0003)	(0.0003)
δ_2 **(Index×Post)** ← **causal**	-0.0007 n.s.	$+0.0031^{***}$
SE	(0.0009)	(0.0008)
N obs	**154,874**	**157,677**

B3. Horse Race: All Four AI Exposure Indices

When run individually, AIOE and GPT Exposure both produce significant positive causal estimates. The horse race places multiple indices simultaneously in one regression to test whether they capture the same AI channel or independent channels.

Table B3: Horse Race — All Four Indices, Trend-Break DID, Log Median Wage

Index	**Individual TB-DID**	**Pairwise (AIOE+GPT)**	**Full Horse Race (All 4)**
AIOE (Felten 2021)	$+0.0031^{***}$	$+0.0015$ n.s.	$+0.0013$ n.s.
SE	(0.0008)	(0.0020)	(0.0023)
GPT Exposure (Eloundou 2024)	$+0.0030^{***}$	$+0.0016$ n.s.	$+0.0015$ n.s.
SE	(0.0008)	(0.0020)	(0.0022)
Frey & Osborne (2013)	-0.0016 n.s.	—	-0.0003 n.s.
SE	(0.0009)	—	(0.0010)
Webb AI Index (2020)	$+0.0015$ n.s.	—	$+0.0016$ n.s.
SE	(0.0008)	—	(0.0010)
N obs	**157,677**	**156,646**	**141,568**
FE: Occ×State + Year + State×Year | SE clustered by occ×state			

*** $p < 0.001$ n.s. = not significant. Index correlations: AIOE–GPT $r = 0.90$, AIOE–F&O $r = -0.424$, AIOE–Webb $r \approx 0.05$. Joint insignificance of AIOE and GPT in the pairwise/full specs reflects near-collinearity ($r = 0.90$) — they capture the same AI channel, not independent mechanisms. Loss of significance in joint regressions is a well-known consequence of including nearly collinear regressors and should not be interpreted as the effect disappearing.

C. Sample Robustness

C1. Drop COVID Years (2020–2021)

COVID-19 caused asymmetric shocks: high-AIOE white-collar jobs were largely shielded (remote-compatible); low-exposure service jobs were severely disrupted. This differential shock could distort the pre-trend estimate δ_1. Dropping 2020–2021 restricts the pre-period to 2019 and the post-period to 2022–2024.

Table C1: Drop COVID Years (2020–2021)

	Log Median Wage		**Log Employment**	
	Full (2019–2024)	**Drop COVID**	**Full (2019–2024)**	**Drop COVID**
δ_1 (AIOE×t)	-0.0078^{***}	-0.0077^{***}	$+0.0077^{***}$	$+0.0082^{***}$
SE	(0.0003)	(0.0003)	(0.0009)	(0.0008)
δ_2 **(AIOE×Post)** ← **causal**	$+0.0031^{***}$	$+0.0040^{***}$	-0.0018 n.s.	$+0.0005$ n.s.
SE	(0.0008)	(0.0008)	(0.0026)	(0.0025)
N obs	**157,677**	**105,424**	**157,677**	**105,424**
FE: Occ×State + Year + State×Year | SE clustered by occ×state				

*** $p < 0.001$ n.s. = not significant. Wage effect strengthens from $+0.31\%^{***}$ to $+0.40\%^{***}$ after dropping pandemic years — COVID was attenuating the main finding, not generating it. Employment null holds in both samples.

C2. Exclude Tech Sector (SOC Group 15)

Computer and Mathematical Occupations (SOC group 15) score highest on AIOE and experienced unusual dynamics in 2022–2024. Excluding these 4 SOC codes (829 state×year obs) tests whether the result is concentrated in the tech sector.

Table C2: Exclude Tech Sector (soc2 $\neq$ 15)

*** $p < 0.001$ n.s. = not significant. Causal wage estimate identical to four decimal places ($+0.0031^{***}$ in both). Result is not a tech-sector story — consistent with task-upgrading mechanism predicting gains for routine/manual workers.

C3. Balanced Panel

The main panel is unbalanced: 4,621 of 27,384 occ×state cells are missing in at least one year due to BLS suppression (insufficient sample size). Restricting to 22,763 cells present

	Log Median Wage		Log Employment	
	Full (667 occs)	**Excl. tech (663 occs)**	**Full (667 occs)**	**Excl. tech (663 occs)**
δ_1 (AIOE×t)	-0.0078^{***}	-0.0078^{***}	$+0.0077^{***}$	$+0.0079^{***}$
SE	(0.0003)	(0.0003)	(0.0009)	(0.0009)
δ_2 **(AIOE×Post)** ← **causal**	$+0.0031^{***}$	$+0.0031^{***}$	-0.0018 n.s.	-0.0020 n.s.
SE	(0.0008)	(0.0008)	(0.0026)	(0.0026)
N obs	**157,677**	**156,848**	**157,677**	**156,848**
FE: Occ×State + Year + State×Year | SE clustered by occ×state				

in all 6 years eliminates non-random attrition.

Table C3: Balanced Panel (22,763 occ×state cells, all 6 years)

	Log Median Wage		Log Employment	
	Unbalanced	**Balanced**	**Unbalanced**	**Balanced**
δ_1 (AIOE×t)	-0.0078^{***}	-0.0076^{***}	$+0.0077^{***}$	$+0.0080^{***}$
SE	(0.0003)	(0.0003)	(0.0009)	(0.0009)
δ_2 **(AIOE×Post)** ← **causal**	$+0.0031^{***}$	$+0.0026^{**}$	-0.0018 n.s.	-0.0057^{*}
SE	(0.0008)	(0.0008)	(0.0026)	(0.0025)
N obs	**157,677**	**136,578**	**157,677**	**136,578**
FE: Occ×State + Year + State×Year | SE clustered by occ×state				

*** $p < 0.001$ ** $p < 0.01$ * $p < 0.05$ n.s. = not significant. Yellow: employment δ_2 changes from -0.0018 n.s. to -0.0057^{*} in balanced panel — cells dropping out may slightly mask a real employment effect (-0.57%). Wage floor result holds: $\delta_2 = +0.0026^{**}$.

D. Estimation Assumptions: Two-Way Clustering

Main spec clusters SE by occ×state (one-way). Two-way clustering (Cameron et al. 2011) simultaneously clusters by occupation AND state, accounting for cross-state occupation shocks and cross-occupation state shocks. SE can only expand or stay equal — coefficients are mathematically identical.

Table D1: Two-Way Clustering (Occupation + State)

*** $p < 0.001$ * $p < 0.05$ n.s. = not significant. Yellow: wage δ_2 significance drops from *** to * ($p = 0.014$) under two-way clustering — SE expands from 0.0008 to 0.0012. Result remains significant. Employment null unaffected.

	Log Median Wage		Log Employment	
	One-way (occ×state)	**Two-way (occ+state)**	**One-way (occ×state)**	**Two-way (occ+state)**
δ_1 (AIOE×t)	-0.0078^{***}	-0.0078^{***}	$+0.0077^{***}$	$+0.0077^{***}$
SE	(0.0003)	(0.0006)	(0.0009)	(0.0021)
δ_2 **(AIOE×Post)** ← **causal**	$+0.0031^{***}$	$+0.0031^{*}$	-0.0018 n.s.	-0.0018 n.s.
SE	(0.0008)	(0.0012)	(0.0026)	(0.0044)
N obs	**157,677**	**157,677**	**157,677**	**157,677**
FE: Occ×State + Year + State×Year | SE clustered by occ×state				

E. External Validation

E1. Direct Inequality Test (p90/p10)

The percentile regressions provide indirect evidence of wage compression by comparing δ_2 across five separate regressions. This test uses log(p90/p10) directly as the outcome — a single regression that directly tests whether ChatGPT reduced wage inequality.

Table E1: p90/p10 Wage Inequality — Direct Test

	log(p90/p10) Overall	**log(p90/p50) Top-half**	**log(p50/p10) Bottom-half**
δ_1 (AIOE×t)	$+0.0050^{***}$	$+0.0045^{***}$	$+0.0007^{*}$
SE	(0.0004)	(0.0003)	(0.0003)
δ_2 **(AIOE×Post)** ← **causal**	-0.0054^{***}	-0.0025^{*}	-0.0031^{**}
SE	(0.0013)	(0.0010)	(0.0011)
N obs	**154,155**	**154,155**	**157,677**
FE: Occ×State + Year + State×Year | SE clustered by occ×state			

*** $p < 0.001$ ** $p < 0.01$ * $p < 0.05$. All three inequality ratios compress significantly. Pre-existing trend ($\delta_1 > 0$) confirms widening inequality before ChatGPT — Standard DID would have misattributed this to the treatment.

E2. CPS 72-Month Employment Event Study

This test uses monthly CPS occupation-level employment data (32,985 occ×month observations across 72 months) to validate the null employment result with greater temporal granularity. Pre-ChatGPT bins validate parallel trends; post-ChatGPT bins validate the employment null.

Table E2: CPS 72-Month Employment Event Study

Period	β (AIOE×Period)	SE	p-value
Pre-COVID (Jan 2019 – Dec 2020)	+0.0046 n.s.	(0.0039)	$p = 0.237$
COVID (Jan 2020 – Dec 2021)	+0.0062 n.s.	(0.0041)	$p = 0.135$
Pre-ChatGPT (Jan – Oct 2022)	−0.0009 n.s.	(0.0041)	$p = 0.831$
ChatGPT Launch Nov 2022 (ref.)	0 (ref.)	—	—
Post-ChatGPT Year 1 (Dec 2022 – Nov 2023)	−0.0030 n.s.	(0.0036)	$p = 0.408$
Post-ChatGPT Year 2 (Dec 2023 – Dec 2024)	−0.0034 n.s.	(0.0038)	$p = 0.366$
N obs: 25,114 | 351 occupations | 72 months (Jan 2019 – Dec 2024)			
FE: Occupation + Year-Month | SE clustered by occupation			

n.s. = not significant (desired). All pre-ChatGPT period bins are n.s. (parallel trends validated). Both post-ChatGPT years are n.s. (no structural employment break). Summary DID pooling all 72 months: $\beta = -0.0069^{***}$ — higher power reveals small persistent drift but no structural break. Data: `cps_emp_monthly.csv` + `nem-occcode-cps-crosswalk.xlsx`.

E3. ACS Individual-Level Validation

200,000 individual-level ACS observations provide a directional cross-check on the BLS OEWS floor-raiser finding using a different survey methodology, sampling frame, and wage concept (weekly vs. annual). ACS 2020 not released by Census Bureau due to COVID-19 disruptions.

Table E3: ACS Individual-Level Validation — Directional Floor-Raiser Test

	p10 (lowest)	**p50 (median)**	**p90 (highest)**
δ_1 (AIOE×t)	−0.0037 n.s.	−0.0018 n.s.	−0.0025 n.s.
SE	(0.0088)	(0.0041)	(0.0045)
δ_2 **(AIOE×Post) ← directional**	+0.0348 n.s.	+0.0098 n.s.	+0.0064 n.s.
SE	(0.0310)	(0.0137)	(0.0152)
N obs (occ×year)	**1,648**	**1,648**	**1,648**

n.s. = expected given substantially lower statistical power (1,648 occ×year cells vs. 157,677 in main analysis; no state dimension). Direction confirmed: $\delta_2(\text{p10}) = +0.035$ is 5.4× larger than $\delta_2(\text{p90}) = +0.006$, consistent with floor-raiser. Data: `acs_indiv_small.csv` + `nem-occcode-cps-crosswalk.xlsx`.